

CLASS 9 MATHEMATICS

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EXERCISE SET 4.2

1. Factor completely:

(i) $9x^2 + 24xy + 16y^2$

(ii) $4s^2 + 20st + 25t^2$

(iii) $49x^2 + 28xy + 4y^2$

(iv) $64p^2 + \frac{32}{3}pq + \frac{4}{9}q^2$

(v) $3a^2 + 4ab + \frac{4}{3}b^2$

(vi) $\frac{9}{5}s^2 + 6sv + 5v^2$

(i)

$$9x^2 + 24xy + 16y^2$$

We know that

$$(a + b)^2 = a^2 + 2ab + b^2.$$

Here,

$$9x^2 = (3x)^2, \quad 16y^2 = (4y)^2$$

and

$$2(3x)(4y) = 24xy.$$

Therefore,

$$9x^2 + 24xy + 16y^2 = (3x + 4y)^2.$$

Hence,

$$\boxed{(3x + 4y)^2}$$

(ii)

$$4s^2 + 20st + 25t^2$$

We have

$$4s^2 = (2s)^2, \quad 25t^2 = (5t)^2$$

and

$$2(2s)(5t) = 20st.$$

Therefore,

$$4s^2 + 20st + 25t^2 = (2s + 5t)^2.$$

Hence,

$$\boxed{(2s + 5t)^2}$$

(iii)

$$49x^2 + 28xy + 4y^2$$

Since

$$49x^2 = (7x)^2, \quad 4y^2 = (2y)^2$$

and

$$2(7x)(2y) = 28xy,$$

we get

$$49x^2 + 28xy + 4y^2 = (7x + 2y)^2.$$

Therefore,

$$\boxed{(7x + 2y)^2}$$

(iv)

$$64p^2 + \frac{32}{3}pq + \frac{4}{9}q^2$$

Observe that

$$64p^2 = (8p)^2$$

and

$$\frac{4}{9}q^2 = \left(\frac{2q}{3}\right)^2.$$

Also,

$$2(8p)\left(\frac{2q}{3}\right) = \frac{32}{3}pq.$$

Therefore,

$$64p^2 + \frac{32}{3}pq + \frac{4}{9}q^2 = \left(8p + \frac{2q}{3}\right)^2.$$

Hence,

$$\boxed{\left(8p + \frac{2q}{3}\right)^2}$$

(v)

$$3a^2 + 4ab + \frac{4}{3}b^2$$

First take 3 as a common factor:

$$3a^2 + 4ab + \frac{4}{3}b^2 = 3\left(a^2 + \frac{4}{3}ab + \frac{4}{9}b^2\right).$$

Now,

$$a^2 + \frac{4}{3}ab + \frac{4}{9}b^2 = \left(a + \frac{2b}{3}\right)^2.$$

Therefore,

$$3a^2 + 4ab + \frac{4}{3}b^2 = 3\left(a + \frac{2b}{3}\right)^2.$$

Hence,

$$\boxed{3\left(a + \frac{2b}{3}\right)^2}$$

(vi)

$$\frac{9}{5}s^2 + 6sv + 5v^2$$

Take 5 as a common factor:

$$\frac{9}{5}s^2 + 6sv + 5v^2 = 5 \left(\frac{9}{25}s^2 + \frac{6}{5}sv + v^2 \right).$$

Now,

$$\frac{9}{25}s^2 + \frac{6}{5}sv + v^2 = \left(v + \frac{3s}{5} \right)^2.$$

Therefore,

$$\frac{9}{5}s^2 + 6sv + 5v^2 = 5 \left(v + \frac{3s}{5} \right)^2.$$

Hence,

$$\boxed{5 \left(v + \frac{3s}{5} \right)^2}$$

2. Find the values of the following using the identity

$$(a - b)^2 = a^2 - 2ab + b^2.$$

(i) $(79)^2$

(ii) $(193)^2$

(iii) $(299)^2$

(i)

$$79^2 = (80 - 1)^2$$

Using

$$(a - b)^2 = a^2 - 2ab + b^2,$$

we get

$$79^2 = 80^2 - 2(80)(1) + 1^2$$

$$= 6400 - 160 + 1$$

$$= 6241.$$

Therefore,

$$\boxed{79^2 = 6241}$$

(ii)

$$193^2 = (200 - 7)^2$$

Using

$$(a - b)^2 = a^2 - 2ab + b^2,$$

$$193^2 = 200^2 - 2(200)(7) + 7^2$$

$$= 40000 - 2800 + 49$$

$$= 37249.$$

Therefore,

$$\boxed{193^2 = 37249}$$

(iii)

$$299^2 = (300 - 1)^2$$

Using

$$(a - b)^2 = a^2 - 2ab + b^2,$$

$$299^2 = 300^2 - 2(300)(1) + 1^2$$

$$= 90000 - 600 + 1$$

$$= 89401.$$

Therefore,

$$\boxed{299^2 = 89401}$$

IMPORTANT IDENTITIES

$$\boxed{(a + b)^2 = a^2 + 2ab + b^2}$$

$$\boxed{(a - b)^2 = a^2 - 2ab + b^2}$$